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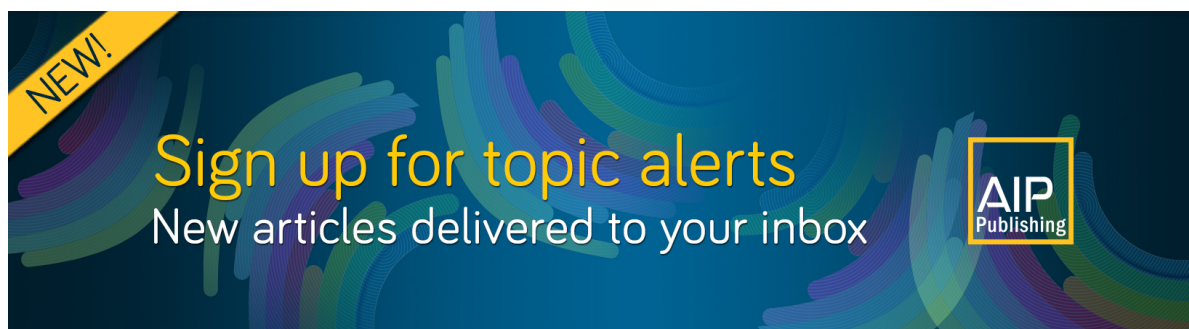
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ABSTRACT

Hydrodynamics characterization and analysis is an essential part in studying mineral dissolution in porous media with complex heterogeneous pore structures including embedded cavities. Cavities affect the pore-scale pressure and flow distribution in the surrounding porous matrix. Transport of the dissolved solute, concentration gradient, and thermodynamic driving forces in that area will be affected as a result of local flow features. Given the properties of cavities and porous media, vorticities may form, and the cavity may partially or fully contribute to the overall flow. Depending on the shape and alignment of the cavity with respect to the direction of general flow, fluid flow will be focused at certain locations on the cavity boundary. Reaction hotspots can form as a result of the facilitated mineral dissolution at those locations. A rigorous flow modeling approach that preserves the flow features inside the cavity and in the porous matrix is used. Stokes flow and seepage flow are applied as two different physics governing the fluid flow in a fluid-filled cavity and a highly permeable sediment-filled cavity consecutively. The analytical model framework permits capturing the detailed flow structure of a single-phase fluid at the curved interface of a prolate spheroidal cavity. The solutions for flow are used within a fully coupled, fully implicit reactive transport simulator to investigate the mineral dissolution in the porous host matrix. The cavity aspect ratio and slip parameter at the border are investigated as the two parameters that affect the dissolution. The simulation results showed that the reaction hotspots are mainly located on the border of the cavity where the influent enters and leaves the cavity. The midpoint between them is where the minimum mineral dissolution was placed. Approximating the cavity as a highly permeable sediment-filled porous zone showed a higher effective reaction rate compared to the fluid-filled cavity. The cavity aspect ratio showed to have a significant impact on the effective reaction rate of the investigated cases. The cavities with a shape closer to a sphere show a higher effective reaction rate.

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I. INTRODUCTION

Chemical dissolution of minerals is known to alter the hydrodynamical and mechanical properties of porous media. Its applications include CO₂ geologic sequestration, hydrocarbon recovery, nuclear waste disposal, acidizing in stimulation of carbonate formations, and contaminant remediation in groundwater.^{1–12} Mineral dissolution can alter the rock properties and affect the fluid flow in the formation, and the local flow regime can, in turn, affect the dissolution rate and change the location of reaction hotspots. These

coupled processes can result in the formation of preferential flow pathways. Depending on the relative magnitude of the solvent velocity and reaction rate, the nonzero solvent velocity would maintain the solute gradient via both advective and dispersive mechanisms and would control the in-place thermodynamic driven forces. Therefore, a proper description of flow is essential for better understanding of the dissolution dynamics and the formation of property alteration in a reactive transport study.

Fluid flow modeling and reactive transport modeling are challenging in porous structures with a multi-modal pore size

distribution, in particular, in natural settings such as carbonate formations, where vugular cavities are scattered throughout the porous host medium.

A cavity, also called a vug, is a void pore with the pore volume much larger than the typical inter-granular pore volume. Cavity diameter may vary from few millimeters to few centimeters. The cavities of interest in this work belong to this size range. Cavities with a diameter of several meters are called caves. In a vugular porous medium, cavities are scattered throughout the porous host matrix and contribute to the overall porosity and permeability of the medium. The fluid flow in the porous matrix near a cavity is significantly affected by the presence of the cavity. As a result, it is expected to have the dissolution being influenced in the area of the porous medium near the cavity. The aim of this work is to investigate how dissolution rates and patterns in a porous medium with scattered cavities are affected by the presence of cavities.

For the simulation of flow in vuggy carbonate formations, where the pore space is composed of the porosity in the porous medium, fractures, and vugs, it is a common practice to apply the multi-continuum approach, e.g., dual porosity or triple porosity models. In this approach, the network of vuggy cavities is considered as a secondary/tertiary porosity system with unitary porosity and high permeability, in which flow obeys Darcy's law.^{13–24} In a recent work, Lei *et al.*²⁵ presented an analytical model for flow in acidized fractured-vuggy media based on a composite domain including the acidized region where the complicated wormholes are treated as a fractal medium and the outer region is treated as a common triple porosity medium. Discrepancies between dual porosity or triple porosity models and pressure transient well-test data are reported in the literature.^{26–31} One of the explanations is the incorrect modeling of cavities in the pressure transient well-test data.

Applying Darcy's law to model flow in a cavity as a highly permeable porous medium is valid only for sediment-filled cavities, also known as porous inclusions, and not for fluid-filled cavities. Darcy's law describes the average of seepage velocity at the scale of representative elementary volume (REV), where the flux (rate) is linearly proportional to the pressure gradient.^{32,33} Darcy's law is inapplicable to nonlinear flows that can arise in cavity media where the cavity volume is expected to be several orders of magnitude larger than the pore size of the porous media. Nevertheless, the multi-continuum models are essentially based on this idea that cavity media co-exist with porous media at every spatial point and are evenly distributed in the medium, which is not the case for the majority of vugular media, e.g., carbonate formations or composite membranes.

Furthermore, in the multi-continuum approach, it is assumed that the whole cavity volume contributes to the overall flow. However, a fluid-filled cavity may only partially contribute to the overall flow due to the formation of vorticities in the cavity. Application of Darcy's law to the cavity cannot capture these features of flow in the cavity. As illustrated in Fig. 1, the streamlines in a highly permeable porous inclusion form a set of straight lines parallel to the flow general direction. The entire inclusion volume is open to fluid flow. However, in a fluid-filled cavity, the streamlines bend toward the cavity centerline. Depending on the porous medium–cavity properties and the cavity morphology, local vortices may form. This means that a part of the cavity volume will not contribute to the overall flow, and the cavity may exhibit a partially

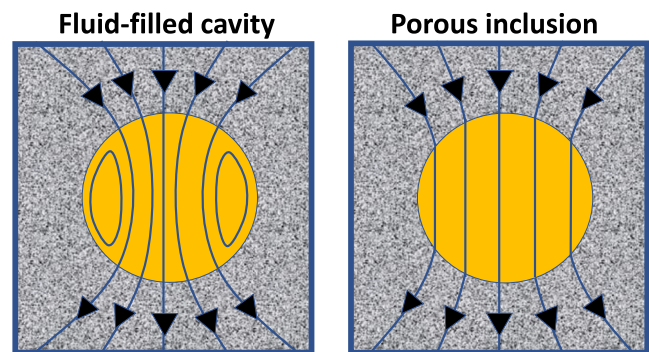


FIG. 1. Schematic of the different fluid flow arrangement in a fluid-filled cavity and in a porous inclusion both embedded in a porous matrix.

no-flow medium in the cavity with lower equivalent permeability than expected.

Another drawback of applying the multi-continuum approach is neglecting the lateral anisotropy in the cavity. The lateral and longitudinal effective permeability of an elongated cavity are different from each other. Moreover, in a multi-continuum framework, the fluid exchange between the cavity and the matrix is considered to be linearly proportional to the difference in the pressures. This implies the continuous pressure and flux (pressure gradient) on the interface. In fact, the condition on the interface of the cavity and the surrounding porous medium is more complicated.

A rigorous modeling approach should preserve the free flow features inside the cavity and treat the interface between the cavity and the matrix properly. Since the size of the cavity is orders of magnitude larger than the pore size and the flow takes place at a relatively low Reynolds number, the Stokes flow is a proper choice to model the flow in the cavity. Imposing the flow coupling boundary condition on the interface between Stokes flow, which is governed by a second-order differential equation, and Darcy flow is difficult. The coupling boundary leads to discontinuity or slip at the interface. At the interface of the cavity and matrix, the normal component of the velocity is continuous, and the tangential velocity possesses a jump condition. The magnitude of this jump, or the so-called slip, depends on the velocity in the cavity as well as the permeability of the porous medium. The Beavers–Joseph–Saffman boundary condition^{34,35} is commonly used to describe the slip of tangential velocity at the interface of Darcy flow and Stokes flow. Moreover, at the interface, the normal stress in fluid will be continuous, which is different from the pressure being continuous.

Numerical modeling of the coupled Darcy flow in the matrix and Stokes flow in the cavity^{36–50} can be challenging, given the complex set of aforementioned boundary conditions at the interface of an irregular shaped cavity. Depending on the accuracy of the discretization and the numerical scheme, mass conservation is not guaranteed.

For a high cavity density, i.e., in a binary medium with a periodic network of cavities evenly distributed in the porous matrix, the homogenization technique can be applied to obtain the effective flow model together with effective porosity and permeability

of the medium.^{51–57} The homogenized model implies the application of Darcy's law for fluid flow in the macro-scale porous medium, which results from the averaged micro-scale flow in one cell of the periodicity of the medium. The cell problem, in general, should be solved numerically for different cavity shapes.⁵¹ This is a coupled Darcy–Stokes flow system that shall be considered. Arbogast and Lehr⁵¹ considered Darcy–Stokes flow in the unit cell and used the complete set of modified Beavers–Joseph–Saffman boundary conditions, which take into account the curved boundary of the cavity. Analytical solutions for the cell problem of a porous cell with embedded spheroidal voids for Darcy–Stokes flow were obtained by Refs. 54 and 55 for a simplified boundary condition of continuous fluid pressure and velocity at the interface of the porous medium and the cavity.

Several authors use the Darcy–Brinkman model to describe a single flow model and smoothly shift from the Darcy flow to Stokes flow by tuning the model parameters.^{58–64} The advantage of this approach is its simplicity, and there is no need to model the special boundary condition at the interface of the cavity and porous medium. Popov *et al.*^{65,66} proposed the homogenized model for the Darcy–Brinkman equation to compute the upscaled effective permeability. In their work, the cell problem was solved numerically using a finite element solver. This approach has also been used to numerically investigate reactive transport processes in heterogeneous porous systems.^{67–74} However, the Darcy–Brinkman model fails in capturing the structure of the flow at the cavity walls, i.e., the slip of tangential velocity component and the jump on pressure, which significantly affects the cavity deformation due to dissolution.

Study of reactive transport in a vugular porous medium is of particular interest in the acidizing process and for CO₂ storage in carbonate formations such as depleted gas reservoirs and karst aquifers.^{75–84,100} The numerical investigation of Izgec *et al.* on CO₂ storage in carbonate formations builds updating the effective porosity and effective permeability of the domain as a function of dissolved rock minerals.⁸⁵ Yang *et al.*⁸³ showed experimentally that the dissolution patterns are drastically different in a homogeneous, fractured, and vuggy limestone flooded with acidic brine. Preferential channels surrounded by branched channels were formed in the homogeneous case, while fractures became the main flow path in the fractured sample. In contrast, only one dominant channel formed in the vuggy sample, which resulted in a sharp permeability increase. They conclude that increasing complexity of the pore geometry leads to more nonuniform dissolution. Izgec *et al.*⁷⁹ observed that acid spreads in wormholes through vuggy rocks faster than those in homogeneous rocks and suggested that the local pressure drop created by vugs was more important in wormhole flow path formation than the chemical reaction at the pore scale. Most of the aforementioned numerical investigations of dissolution models^{76,78–84} are based on some sort of simplifying flow modeling, i.e., effective medium, multi-continuum approach, or Stokes–Brinkman approach.

The objectives of this work are to employ a more accurate description of flow in the vicinity of the fluid-filled cavity and look into its impacts on mineral dissolution. Capturing these processes are also needed for investigation of cavity deformation. For this purpose, we use a rigorous formulation of flow and utilize analytical solutions for the steady-state coupled Darcy–Stokes flow. The analytical solution for a 2D Darcy–Stokes flow in a porous medium

including a single circular or slightly deformed circular-shaped cavity was obtained by Raja Sekhar and Sano.^{86,87} They used a simplified set of boundary conditions of the continuity of the velocity and stress components on the cavity surface. Markov *et al.*⁸⁸ presented the analytical solution for the fluid flow and effective permeability of porous media with embedded circular and spherical fluid-filled cavities. They coupled Stokes flow and Darcy flow through a set of Beavers–Joseph boundary conditions with the assumption of a flat interface between the free flow and the porous medium. We use the set of analytical solutions for steady-state coupled Darcy–Stokes flow with modified Beavers–Joseph–Saffman boundary conditions presented in Ref. 89. Overall, this analytical solution considers flow discontinuity on the curved interface of the cavity and thus permits a more accurate description of the local flow structures. The analytical results for the fluid velocity are used in a fully coupled, fully implicit reactive transport simulator⁹⁰ to model the solute transport and mineral dissolution.

This study compares two different approaches in modeling flow in a porous medium with embedded cavities: approximating a cavity as a highly permeable porous zone vs modeling the cavity as a fluid-filled zone. We claim that the analytical solution used for flow modeling in this work captures the flow features better, and as a result, the reactive transport using this flow model will be more accurate. We compare the results of these two approaches and show how the flow field and dissolution pattern can be different when approximating the cavity as a highly permeable porous region compared to the case of a fluid-filled zone.

This paper is organized as follows: In Sec. II, the governing equations for flow and transport are provided for two cases of the fluid-filled cavity and a highly permeable porous inclusion embedded at the center of a porous matrix. The formulation for solute transport and geochemical dissolution is provided in Sec. III. Results and discussion are provided in Sec. IV, and finally, conclusions are provided in Sec. V. Most of the mathematical derivations are provided in the appendices.

II. MATHEMATICAL FORMULATION OF THE PROBLEM

A. Governing equations of flow

The steady-state flow of an incompressible single-phase fluid in a fully saturated axis-symmetric porous medium, Ω_m , with an embedded fluid-filled cavity, Ω_c , located at the center is considered (Fig. 2). $\Gamma_{m,c}$ is the interface of the porous matrix and the cavity, and $\partial\Omega$ denotes the external boundary. The right-handed Cartesian coordinate system is set with the x axis in the direction of the elongated cavity axis and y axis normal to the plane pointing outward. The origin of the coordinate system is located at the center of the cavity. This choice of the axes of the coordinate system is for further convenience in deriving solutions for flow. The medium is exposed to an external uniform pressure gradient in the x -direction. The size of a matrix cell is assumed to be sufficiently large compared with the cavity size; therefore, the far-field pressure remains unaffected by the presence of the cavity.

In the following, the subscript m refers to the matrix variables and properties, and the subscript c refers to the cavity variables and properties. The fluid properties are assumed to be the same in the matrix and cavity and do not change when dissolution takes place.

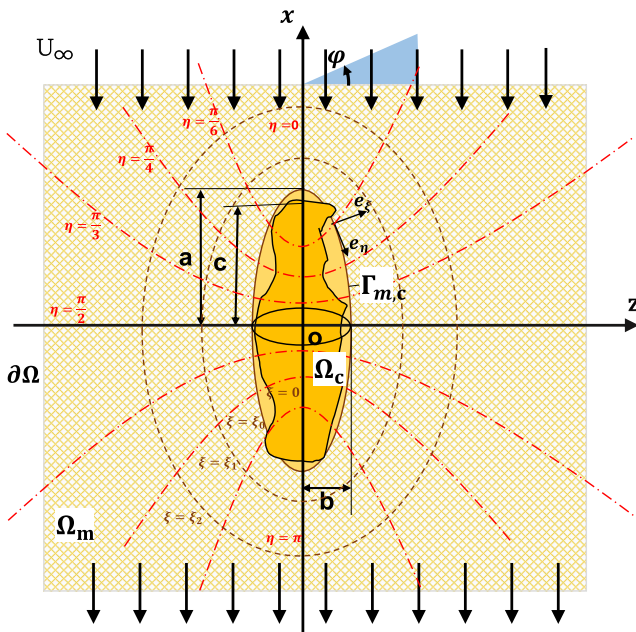


FIG. 2. General schematic of the problem of flow in a porous medium with an embedded cavity at the center approximated by a spheroid with a semi-major axis and semi-minor axis of length a , b and focal length c ; the relation between the Cartesian coordinate system (x, y, z) and the spheroidal coordinate system (ξ, η, ϕ) is given in Appendix A.

The seepage flow in the porous matrix Ω_m is governed by Darcy's law. Mass and momentum conservation can be expressed as

$$\nabla \cdot \mathbf{u}_m = 0 \quad \text{in } \Omega_m, \quad (1)$$

$$\mathbf{u}_m = -\frac{k_m}{\mu} \nabla p_m \quad \text{in } \Omega_m, \quad (2)$$

where $\mathbf{u}_m = \langle u_m, v_m \rangle$, p_m , k_m , and μ denote the velocity vector, pressure, and permeability in the matrix medium, and the fluid dynamic viscosity, respectively. The stationary pressure field in the matrix from Eqs. (1) and (2) holds the following equation:

$$\nabla^2 p_m = 0 \quad \text{in } \Omega_m. \quad (3)$$

Inside the cavity, the fluid flow is assumed to be laminar at a small Reynolds number with negligible inertial forces, which is described by Stokes flow. Thus, the steady-state mass and momentum balance equations can be written as

$$\nabla \cdot \mathbf{u}_c = 0 \quad \text{in } \Omega_c, \quad (4)$$

$$\nabla p_c + \mu \nabla \times \nabla \times \mathbf{u}_c = 0 \quad \text{in } \Omega_c, \quad (5)$$

where $\mathbf{u}_c = \langle u_c, v_c \rangle$ and p_c denote the velocity vector and the pressure in the cavity, respectively. All the differential operators are given in Appendix A for a general orthogonal curvilinear coordinate system.

The problem is closed with the following set of regularity and boundary conditions on the cavity and matrix interface:

Bounded velocity at the cavity center:

$$\lim_{r \rightarrow 0} |u_c| < \infty, \quad \lim_{r \rightarrow 0} |v_c| < \infty, \quad (6)$$

where r is the distance of any point inside the cavity from the origin of the coordinate system.

Regularity of the pressure field at infinity:

$$\lim_{x \rightarrow \infty} p_m = |\nabla(p_m)_\infty| x. \quad (7)$$

Here, $\nabla(p_m)_\infty$ is the uniform pressure gradient in the matrix in the x -direction at infinity and is not perturbed due to the presence of the cavity.

Continuity of the normal velocities on the interface: This guarantees the conservation of mass between the matrix and the cavity,

$$\mathbf{u}_c \cdot \mathbf{n} = \mathbf{u}_m \cdot \mathbf{n} \quad \text{on } \Gamma_{m,c}, \quad (8)$$

where $\mathbf{n}$ is the unit normal to the interface.

Pressure jump on the interface: It represents the balance of two driving forces in the normal direction along with the interface, the kinematic pressure in the matrix, and the normal component of the normal stress in the cavity,

$$-p_m = -p_c + 2\mu \mathbf{n} \cdot \boldsymbol{\epsilon} \cdot \mathbf{n} \quad \text{on } \Gamma_{m,c}, \quad (9)$$

where $\boldsymbol{\epsilon} = \frac{1}{2}(\nabla \mathbf{u}_c + \nabla \mathbf{u}_c^T)$ is the rate of strain tensor.

The slip condition for tangential velocities on the interface: this condition addresses how the matrix affects the flow at the interface. Beavers and Joseph³⁴ showed experimentally that a free fluid in contact with a porous medium flows faster than a fluid in contact with a solid surface. This observation is taken into account by imposing a slip boundary condition on the interface of the matrix and free fluid contact. Later, Ref. 35 showed that this slip is independent from the tangential velocity in the porous media. In this work, we use the generalized form of the Beavers–Joseph–Saffman slip boundary condition proposed by Ref. 91 for complicated interface geometries other than the plane interface,

$$\mathbf{n} \cdot \boldsymbol{\epsilon} \cdot \boldsymbol{\tau}_i = \frac{\lambda}{\sqrt{\boldsymbol{\tau}_i \cdot k_m \cdot \boldsymbol{\tau}_i}} \mathbf{u}_c \cdot \boldsymbol{\tau}_i, \quad i = 1, 2, \quad \text{on } \Gamma_{m,c}, \quad (10)$$

where k_m is the matrix permeability, $\boldsymbol{\tau}_i$ is the tangential unit vector to the interface (Fig. 2), and λ is an empirical coefficient called the slip parameter or Beavers–Joseph constant, which depends on the flow type, thickness of the free fluid boundary layer, flow direction, and position of the interface.^{92–94} For axis-symmetric problems, $i = 1$ in the slip boundary condition.

B. Prolate spheroidal cavity

The shape of a cavity is approximated by a prolate spheroid. Spheroidal forms (axis-symmetric ellipsoids) are ideal choices, and they can be used for addressing a wide variety of geometrical shapes. A spheroid reduces to a sphere when the focal length of the spheroid approaches zero. An oblate spheroid degenerates to a flat disk that can represent a fracture geometry. A prolate spheroid degenerates to a long rod that can represent a connected cavity geometry when the focal length of the spheroid approaches infinity. We consider a prolate spheroidal fluid-filled cavity, Ω_c , located in the homogeneous

matrix medium Ω_m with major axis a in the x -direction and minor axis b in the y, z directions with $a > b$, as shown in Fig. 2.

The prolate spheroidal coordinate system with the coordinates (s, t, ϕ) is given in Appendix B. The set of points inside the spheroidal cavity are given by

$$\Omega_c = \{(s, t, \phi) : 1 \leq s < s_o, -1 \leq t \leq 1, 0 \leq \phi \leq 2\pi\}, \quad (11)$$

and the points in the porous matrix are given by

$$\Omega_m = \{(s, t, \phi) : s_o < s, -1 \leq t \leq 1, 0 \leq \phi \leq 2\pi\}, \quad (12)$$

with $s_o = a/c$, which defines the prolate spheroid of semi-axes a, b as the surroundings of the cavity.

C. Analytical solution for flow in the cavity and porous matrix

The stream function inside the spheroidal cavity is given as (see Appendix C)

$$\Psi_c(s, t) = c^2 \sum_{k=1}^{\infty} g_{1,2k}(s) G_{2k}(t). \quad (13)$$

Here, G_n and H_n are the Gegenbauer polynomials of first and second kind and of degree $-\frac{1}{2}$; g_n is given in Eq. (C13). Defining the stream function in the spheroidal cavity, the velocity field in the cavity can be obtained using Eq. (C1). Consequently, the pressure field inside the cavity using Eq. (C3) is derived as

$$p_c(s, t) = - \sum_{k=1}^{\infty} \frac{\mu c}{2k(2k-1)} C_{1,2k} P_{2k-1}(s) P_{2k-1}(t), \quad (14)$$

where $P_n(t)$ is the Legendre polynomial of first kind and order n and $C_{1,2k}$ is the undetermined coefficient in the series expansion to be defined later.

The pressure field in the porous matrix, outside the cavity, satisfies Eq. (3), with the Laplacian differential operator for the spheroidal coordinate system given in Eq. (B10). Considering the regularity of the far-field pressure in Eq. (7) as

$$\lim_{s \rightarrow \infty} p_m(s, t) = |\nabla(p_m)_{\infty}| c s t, \quad (15)$$

the pressure field outside the spheroidal cavity is given by

$$p_m(s, t) = \sum_{k=1}^{\infty} c F_{2k-1}(s) P_{2k-1}(t), \quad (16)$$

with

$$F_{2k-1}(s) = |\nabla(p_m)_{\infty}| s \delta_{1,2k-1} + A_{2k} Q_{2k-1}(s), \quad (17)$$

where $\delta_{1,2k-1}$ is the Kronecker delta function, $Q_n(s)$ is the Legendre polynomial of second kind and order n , and A_{2k} are the coefficients to be determined later. As the pressure field has to be odd with respect to the axis of symmetry, only odd terms are included in this solution. Note that the pressure field holds the regularity condition when $s \rightarrow \infty$ as

$$\lim_{s \rightarrow \infty} \frac{Q_{2k-1}(s) P_{2k-1}(t)}{c s t} = 0. \quad (18)$$

Darcy's law gives the velocity field in the matrix by Eq. (2). The stream function in the matrix yields

$$\Psi_m(s, t) = - \frac{c^2 k_m (s^2 - 1)(1 - t^2)}{\mu 2k(2k-1)} F'_{2k-1}(s) P'_{2k-1}(t). \quad (19)$$

Here ' denotes the derivative. This solution holds the condition of a constant far-field pressure gradient as

$$\lim_{x \rightarrow \infty} \Psi_m = \frac{1}{2} U_{\infty} r^2, \quad (20)$$

with $r^2 = c^2(s^2 - 1)(1 - t^2)$ and $U_{\infty} = -\frac{k_m}{\mu} |\nabla(p_m)_{\infty}|$.

D. Coupling flow in the matrix and cavity

On the interface of the cavity and matrix where $s = s_o$, the boundary conditions given by Eqs. (8)–(10) in spheroidal coordinates are applied to find the undetermined coefficients $C_{1,2k}$, $a_{1,2k}$, A_{2k} . Having $3N$ equations given by Eqs. (8)–(10), we can obtain $3N$ unknowns $C_{1,2k}$, $a_{1,2k}$, A_{2k} by solving a linear algebraic system of equations. To match coefficients of different Legendre or Gegenbauer polynomials, all of the terms are written in the series expansion of the Legendre polynomials. The infinite sum $\sum_{k=1}^{\infty}$ is replaced by a finite sum $\sum_{k=1}^N$ with $N \geq 1$ in the computational implementation. The coupling flow problem in the cavity and matrix is given in Appendix D.

E. Flow in the medium with an embedded porous inclusion

In addition to fluid-filled cavities, we also investigate highly permeable porous inclusions. The steady-state flow in a porous medium, Ω_m , with a highly permeable inclusion of spheroidal shape Ω_{in} located at its center and with permeability k_{in} and unitary porosity is described by Darcy's law. Combining Darcy's law with the law of conservation of mass, the pressure fields in the matrix and cavity can be expressed as

$$\nabla^2 p_m = 0 \quad \text{in } \Omega_m, \quad (21)$$

$$\nabla^2 p_{in} = 0 \quad \text{in } \Omega_{in}. \quad (22)$$

Here, the subscript *in* refers to the highly permeable inclusion. Flow in the inclusion is coupled to the flow in the matrix through the following set of no jump on pressure and flux boundary conditions on the inclusion border $\Gamma_{m,in}$,

$$\begin{aligned} p_{in} &= p_m & \text{on } \Gamma_{m,in}, \\ \frac{k_{in}}{\mu} \frac{\partial p_{in}}{\partial s} &= \frac{k_m}{\mu} \frac{\partial p_m}{\partial s} & \text{on } \Gamma_{m,in}. \end{aligned} \quad (23)$$

Matrix pressure, similar to the case of a porous medium with a fluid-filled cavity, is given in Eq. (16). The condition of regularity of the pressure at the inclusion center yields the solution to Eq. (22) as

$$p_{in}(s, t) = c \sum_{k=1}^{\infty} D_{2k-1} P_{2k-1}(s) P_{2k-1}(t), \quad (24)$$

where D_{2k-1} is a constant to be determined using the boundary conditions for coupling flow in the matrix and in the inclusion [Eq. (23)]. The following system of $2N$ equations, for $N \geq 1$, should be solved to define the $2N$ unknowns D_{2k-1} and A_{2k} ,

$$\begin{aligned} \sum_{k=1}^{\infty} D_{2k-1} P_{2k-1}(s_o) - \sum_{k=1}^{\infty} A_{2k} F_{2k-1}(s_o) &= 0, \\ k_{in} \sum_{k=1}^{\infty} D_{2k-1} P'_{2k-1}(s_o) - k_m \sum_{k=1}^{\infty} A_{2k} F'_{2k-1}(s_o) &= 0. \end{aligned} \quad (25)$$

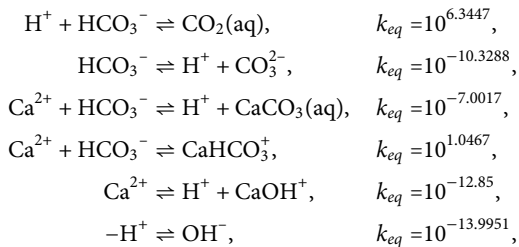
III. FULLY COUPLED REACTIVE TRANSPORT

A. Geochemical reaction modeling

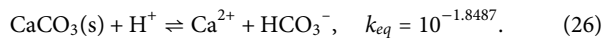
Dissolution of a single mineral, calcite, is investigated. Calcite is a fast-reacting mineral that is abundant in subsurface formations. The carbonate-rich (calcite mineral fraction of about 50%) solid phase is completely contained within the porous matrix. Initially, the medium is saturated with brine of pH = 6.83 and the low concentration of the aqueous calcium ion $\text{Ca}^{2+}(\text{aq})$. Although the results of this study are for a single mineral dissolution, the proposed modeling framework can be easily adopted for multiple mineral dissolution. 0.05M NaCl brine acidified by CO_2 under ambient pressure conditions and unsaturated with respect to calcite enters the medium from the top boundary. It is transported out of the medium across the bottom boundary. The left and right boundaries are kept at no-flow conditions. The mineral dissolution takes place in the matrix and at the interface of the matrix with the cavity at $s = s_o$. We consider the situation where all aqueous reactions reach equilibrium instantly.

The primary chemical species are Ca^{2+} , H^+ , and HCO_3^- , and the secondary species are $\text{CO}_2(\text{aq})$, CO_3^{2-} , $\text{CaCO}_3(\text{aq})$, CaHCO_3^+ , CaOH^+ , and OH^- .

The following are the aqueous geochemical reactions taken into account:



and one kinetic reaction to dissolve mineral species CaCO_3 ,



Here, k_{eq} is the equilibrium constant. The specific reactive surface area is $a_m = 0.461 \text{ m}^2/\text{l}$, kinetic rate constant $k_{rm} = 6.456542 \times 10^{-2} \text{ mol/m}^2/\text{s}$, and activation energy $E_a = 15.000 \text{ J mol}^{-1}$.⁹⁵

B. Solute transport modeling

The problem consists of a complicated chemical reaction network under advection and diffusion.

The mass conservation equation for the total molar concentration C_i of each primary chemical species has the form⁹⁶

$$\begin{aligned} \frac{\partial C_i}{\partial \tau} + \mathbf{u} \cdot \nabla C_i &= D \nabla^2 C_i + R_{m,i} \quad \text{in } \Omega_m \cup \Omega_c, \\ i &= \text{Ca}^{2+}, \text{H}^+, \text{HCO}_3^-, \text{CO}_3^{2-}, \end{aligned} \quad (27)$$

where τ is time; D is the molecular diffusion coefficient, which is considered to be the same for all the species; and $\mathbf{u} = \mathbf{u}_m$ or $\mathbf{u}_c$

depending on the grid point location. For the advective term, we made use of the incompressibility condition in the matrix and cavity [Eqs. (1) and (4)]. $R_{m,i}$ is the reaction source/sink term. No mineral dissolution takes place in the cavity, and $R_{m,i}$ is nonzero in the matrix and on the matrix–cavity interface,

$$\begin{aligned} R_{m,i} &= \sum_{m=1}^{N_m} v_{im} I_m & \text{in } \Omega_m \text{ and on } \Gamma_{m,c}, \\ R_{m,i} &= 0 & \text{in } \Omega_c. \end{aligned}$$

N_m is the number of minerals formed by kinetic reactions and

$$I_m = k_{rm} a_m \left| 1 - \frac{1}{k_{eq,m}} \Pi_i (\gamma_i C_i)^{v_{im}} \right|, \quad (28)$$

where $k_{eq,m}$ is the equilibrium constant for mineral species, v_{im} is the stoichiometric coefficient in kinetic reactions, and γ_i is the activity coefficient.

The total concentration of the i th primary species, C_i , is defined as

$$C_i = c_i + \sum_{j=1}^{N_{2nd}} v_{ij} c_j, \quad (29)$$

where c_i is the concentration of the primary species, c_j is the concentration of the secondary species, N_{2nd} is the total number of secondary species, and v_{ij} are the stoichiometric coefficients in aqueous reactions.

For aqueous equilibrium reactions, the concentration of the j th secondary species, C_j , is calculated from mass action equations corresponding to equilibrium reactions, which yields

$$C_j = \frac{k_{eq,j}}{\gamma_j} \prod_{i=1}^{N_c} (\gamma_i C_i)^{v_{ij}}, \quad (30)$$

where $k_{eq,j}$ is the equilibrium constant and N_c is the number of primary species.

The rate of change of the molar concentration, C_m , of the m th mineral species due to dissolution is

$$\frac{\partial C_m}{\partial \tau} = I_m. \quad (31)$$

The fully coupled advection–diffusion–reaction equation is solved numerically on a triangular grid that conforms to the geometry of the cavity. A schematic of the grid is shown in Fig. 3. It should be noted that for the sake of computational needs, the grid used in this study is finer than what is schematically shown in this figure. For example, for a spherical cavity geometry, the grid was created using 100 nodes on the cavity interface.

For the coupling scheme, the steady-state solutions for velocity are used within a fully implicit finite element based code for reactive transport modeling by MOOSE (Multiphysics Object-Oriented Simulation Environment).^{90,97,99} First, the flow model is initialized, given the fluid, formation, and cavity properties, including the cavity geometry and the external pressure gradient. Then, the fluid velocity is obtained analytically at the midpoint of element sides. Then, the velocity vector is used in the fully coupled reactive transport module to simulate the reactive transport process of mineral dissolution. The transport module is initialized based on the initial distribution of the

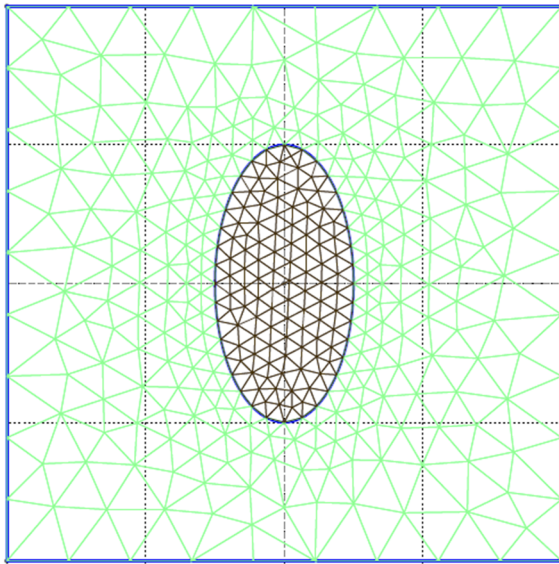


FIG. 3. A schematic of the unstructured triangular grid. It should be noted that for the sake of computational needs, the actual grid used in this study is finer than what is schematically shown in this figure.

mineral species. All simulations are run until the calcium ion concentration at the outlet reaches the steady state and does not change with time.

Two parameters are considered to investigate the local and overall mineral dissolution: (1) The local cumulative reaction source/sink on the cavity surroundings during the numerical simulation can show the local reaction hotspots on the perimeter of the cavity and potential changes to cavity geometry. (2) The average calcium ion (Ca^{2+}) concentration flux at the matrix outlet, same as the one introduced in Ref. 98, is computed to quantify the bulk effective reaction rate as

$$R_{\text{bulk}, i} = \int_0^L c_i(L, z) u(L, z) dz, \quad i = \text{Ca}^{2+}. \quad (32)$$

In this paper, the evolution of the cavity during the dissolution process is not considered, and therefore, the fluid velocity, which depends on the cavity shape, is at the steady-state condition.

IV. RESULTS AND DISCUSSION

A. Flow and mineral dissolution in the porous inclusion and cavity

The results for flow and cumulative mineral dissolution around a prolate spheroidal fluid-filled cavity are compared against the flow and cumulative mineral dissolution around the same cavity when it is approximated as a sediment-filled highly permeable porous inclusion. The porous matrix permeability is set as $k_m = 1$ mD. The inclusion is considered to have unitary porosity and $k_{in} = 100\,000$ mD permeability. Three different aspect ratios for spheroidal geometry are considered: $a \sim b$, $a = 2b$, $a = 4b$. In all numerical simulations, the medium is exposed to the same external pressure field with the same

far-field pressure gradient. The presence of the cavity perturbs the fluid flow near the cavity. The streamlines lean toward the cavity axis line. The slip parameter at the fluid-filled cavity boundary, where free flow meets the seepage flow, determines how the streamlines deviate from their original direction in the porous matrix. Three different values of slip parameter for low, intermediate, and high slip are considered.

In a highly permeable porous inclusion, a no-jump condition for pressure and flux applies to the interface of the inclusion and porous medium. No vorticities can form within the inclusion. The whole inclusion volume contributes to the flow. The streamlines in the porous matrix will lean toward the inclusion centerline as there is less resistance for fluid to pass through. Because the flow in the inclusion obeys Darcy's law, the streamlines in the inclusion remain as straight lines. The results for streamlines in a spherical inclusion and a spheroidal inclusion with $a = 2b$ are presented in Fig. 4(e) (bottom row) and Fig. 5(e) (bottom row).

In a fluid-filled cavity, the streamlines can be nonlinear, and depending on the boundary condition on the cavity border, vorticities may form. Figures 4(f)–4(h) (bottom row) show the streamlines in a spherical fluid-filled cavity and in the surrounding porous medium. As shown in these figures, the more the slip on the cavity–matrix interface, the more the streamlines deviate. For a large slip parameter, vorticities can form [Fig. 4(h) (bottom row)]. Vorticities reduce the volume of the cavity that contributes to the flow.

Similar results are presented for streamlines in and near a spheroidal cavity with $a = 2b$ in Figs. 5(f)–5(h) (bottom row) with a low, medium, and high slip on the interface, consecutively. As can be observed in Fig. 5(g) (bottom row), with an increase in the slip parameter, vorticities arise in the area adjacent to the cavity boundaries, and a small cavity volume, at the centerline, remains open to fluid to pass through. It is seen that with a smaller aspect ratio, vorticities form at lower slip parameter magnitude. For the large slip parameter, it is possible that the whole volume of the cavity remains blocked, and a boundary layer near the walls remains open to flow [Fig. 5(h) (bottom row)].

In Figs. 4(e) and 5(e) (bottom row), the loss in calcite mineral fraction (from the initial content of 50%) is shown for a highly permeable porous spherical/spheroidal inclusion. From the same numerical simulation, the loss of the mineral around a spherical/spheroidal fluid-filled cavity with different slip parameters on the cavity–porous medium interface is presented [Figs. 4(f)–4(h) (bottom row) and Figs. 5(f)–5(h) (bottom row)]. As shown in these figures, in all the cases, most part of the mineral dissolution takes place in the porous matrix around the lower half of the cavity/inclusion near to the cavity/inclusion downstream. The cavity downstream is where the fluid leaves the cavity. Another dissolution hotspot is located at the cavity/inclusion upstream, at $x = -0.5$, where the fluid enters the cavity. The ultimate cavity deformation is expected to result in a more elongated cavity shape. The elongation will be in the direction of general flow. Although these graphs can be helpful in locating the reaction hotspots, not much differentiating information can be extracted on how the slip condition or the cavity internal flow can affect the dissolution process in the porous medium. For the spherical fluid-filled cavity $a = b$, the slip parameter does not affect the dissolution [Figs. 4(f)–4(h) (bottom row)], but for the spheroidal fluid-filled cavity $a = 2b$, there is more dissolution at the

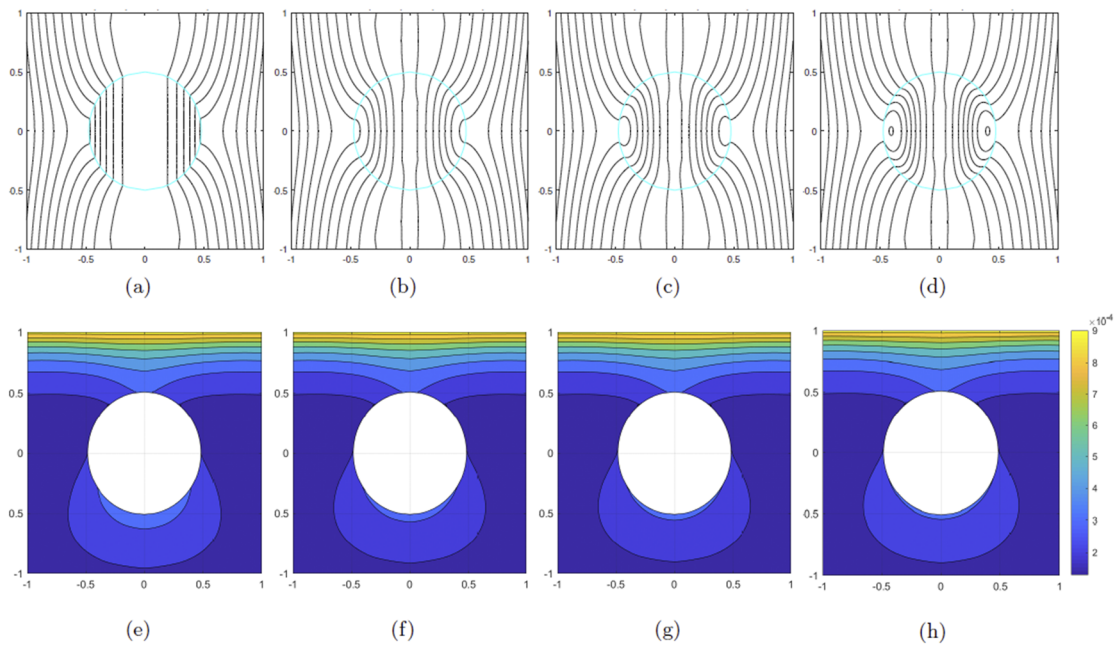


FIG. 4. Streamlines [first row (a)–(d)] and reaction hotspots and loss in calcite mineral fraction [second row (e)–(h)] for a porous inclusion and fluid-filled cavity of $a \sim b$: porous inclusion with no slip [(a) and (e)], fluid-filled cavity with mild slip $\lambda = 1.0$ [(b) and (f)], fluid-filled cavity with moderate slip $\lambda = 5.0$ [(c) and (g)], and fluid-filled cavity with large slip, $\lambda = 8.0$ [(d) and (h)].

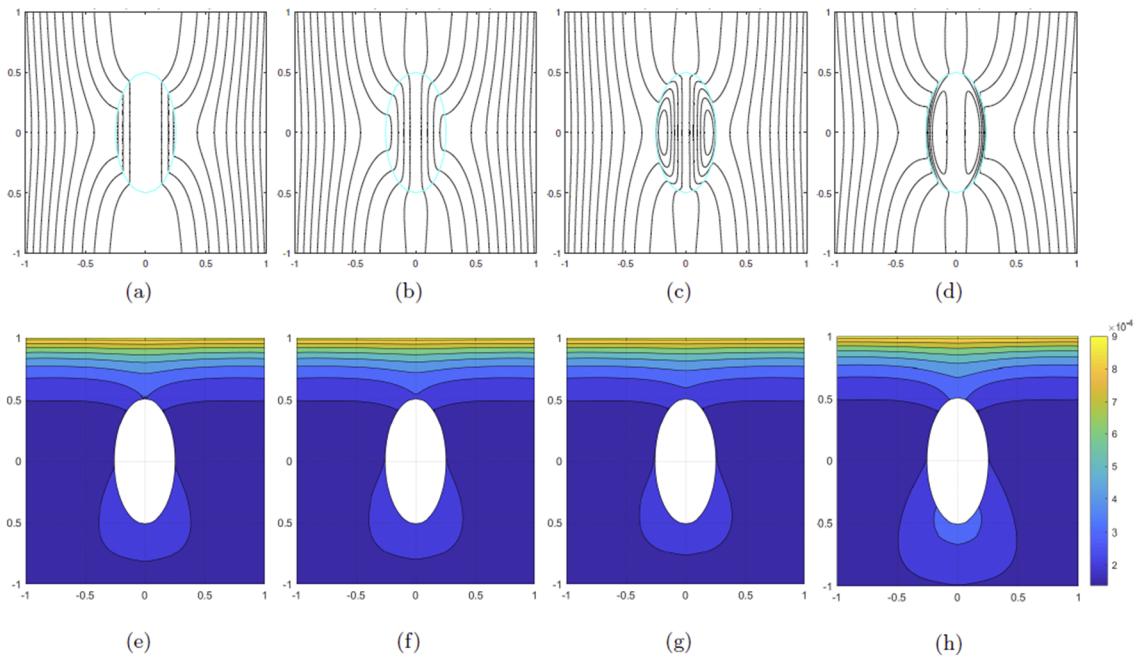


FIG. 5. Streamlines [first row (a)–(d)] and reaction hotspots and loss in calcite mineral fraction [second row (e)–(h)] for a porous inclusion and fluid-filled cavity of $a = 2b$: porous inclusion with no slip [(a) and (e)], fluid-filled cavity with mild slip $\lambda = 1.0$ [(b) and (f)], fluid-filled cavity with moderate slip $\lambda = 5.0$ [(c) and (g)], and fluid-filled cavity with large slip, $\lambda = 8.0$ [(d) and (h)].

downstream when the slip parameter is very large [Figs. 5(f)–5(h) (bottom row)].

One potential comment on Fig. 4 (bottom row) is that the dissolution at the spherical inclusion downstream is slightly more important compared to the dissolution at the spherical cavity downstream. The same comparison between a spheroidal inclusion and a spheroidal cavity with a slightly elongated geometry of $a = 2b$ results in no significant difference [Fig. 5 (bottom row)]. In the elongated shapes with a relatively large slip parameter, the mineral dissolution hotspot at the cavity downstream is more important [Fig. 5(h) (bottom row)] compared to the lower slip parameters [Figs. 5(f) and

5(g) (bottom row)] and dissolution around an inclusion [Fig. 5(e) (bottom row)].

To further investigate how the cavity may deform due to the dissolution of the surrounding porous medium, we look into the dissolution profile on the half of the cavity border starting from the top upstream point on the cavity to the very bottom downstream point on the cavity and then on the porous medium centerline all the way to the medium outlet (Fig. 6). The mineral loss in the porous medium along the border of an elongated spheroidal, $a = 4b$, highly permeable inclusion (solid line) and a fluid-filled cavity of the same geometry with various slip parameters (dashed lines) is shown in

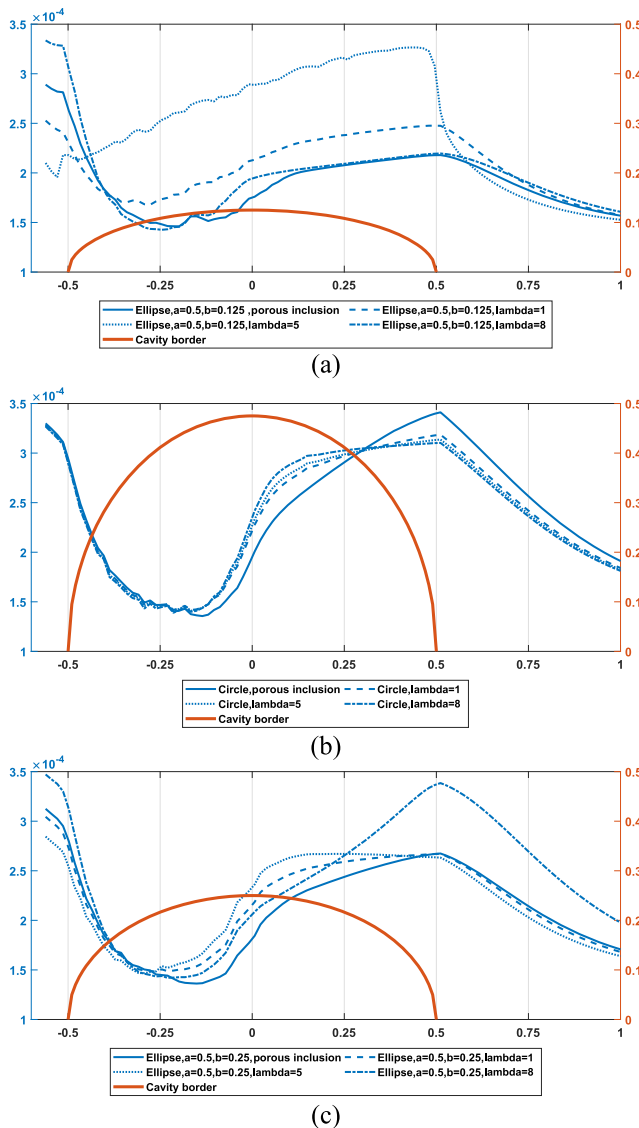


FIG. 6. Cumulative reduction in the calcite mineral fraction in the porous medium around the cavity (blue lines) and cavity border geometry (red line): cavity of $a = 4b$ (a), cavity of $a = 2b$ (b), and cavity of $a \sim b$ (c); the horizontal axis corresponds to the coordinates in the x-direction.

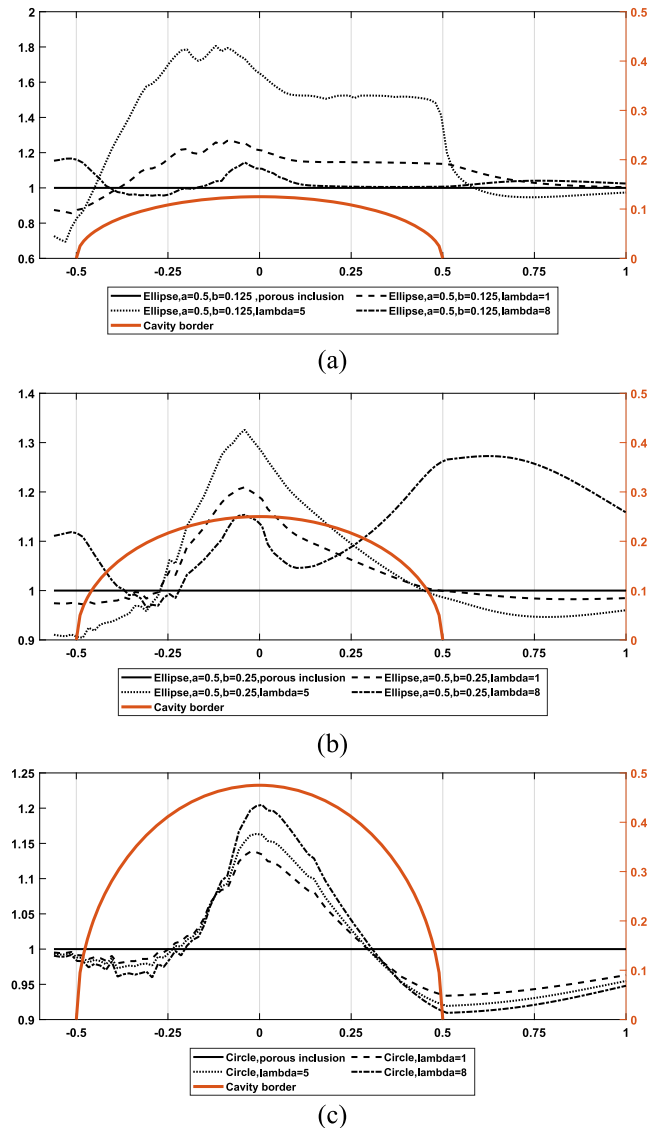


FIG. 7. Reduction in the mineral content around the cavity in the fluid-filled cavity normalized to the reduction in the mineral content around the porous inclusion of the same geometry: cavity of $a = 4b$ (a), cavity of $a = 2b$ (b), and cavity of $a \sim b$ (c); the horizontal axis corresponds to the coordinates in the x-direction.

Fig. 6(a). Only the half right part of the cavity is presented here. The cavity border is plotted in orange to facilitate associating reaction hotspots with the location on the border. For almost all the cases except the large slip parameter, two local maxima at the cavity/inclusion upstream, at $x = -0.5$, and downstream, at $x = 0.5$, are observed. The upper sides of the cavity/inclusion show less reduction in mineral compared to the lower sides. This can be interpreted as the ultimate widening in the half bottom of the cavity/inclusion. In an elongated fluid-filled cavity, the dissolution profile with a larger slip parameter is more similar to that of a porous inclusion. The low and intermediate slip parameters result in more dissolution than that of a porous inclusion along the inclusion border except at the upstream inlet location.

Figure 7(a) compares how the loss of mineral at each point on the elongated spheroidal fluid-filled cavity border, with $a = 4b$, is different from the corresponding point on the porous inclusion of the same geometry. In this figure, the loss in the mineral content at each point on the cavity border is normalized with respect to the loss in the mineral content at the same point on the porous inclusion border. The solid line, for inclusion, is unitary everywhere. As can be seen in Fig. 7(a), the loss of mineral around $x = 0$ on the cavity border of an elongated fluid-filled cavity is at least 10% underestimated when the cavity is approximated by a porous inclusion.

For a slightly larger aspect ratio, $a = 2b$, as in Fig. 6(b), in most part of the lower half of the fluid-filled cavity, $x > 0$, the mineral loss is more than the corresponding point on the porous inclusion of the same size. Here again, the upstream, $x = -0.5$, and downstream, $x = 0.5$, points are the spots on the cavity border where the maximum reaction takes place. As we move away from the cavity, $x \geq 0.5$, the impact of the cavity becomes negligible, and all the curves converge. As can be observed in Fig. 7(b), the mineral fraction loss on the border of a porous inclusion at the most part of the border is less than the loss of mineral on the corresponding fluid-filled cavity. The maximum discrepancy is around the cavity midpoint, at $x = 0$, where dissolution on the inclusion border is at least 15% underestimated compared to a fluid-filled cavity.

For a large aspect ratio of the spheroid, $a \sim b$, one can see that the mineral dissolution in the lower half of the cavity is more than the corresponding case inclusion [Fig. 6(c)], and at points closer to the downstream, $x = 0.5$, the mineral dissolution on the inclusion border will be more. Again, the maximum discrepancy in the dissolution on the cavity border takes place at a midpoint, around $x = 0$, between the cavity upstream and downstream outlet [Fig. 7(c)]. For a large aspect ratio, the slip parameter does not make a significant difference in the dissolution profile along the border of the cavity compared to the one along the inclusion border.

B. Effective reaction rate

To quantify the impact of the cavity on the mineral dissolution rate, the effective reaction rate given in Eq. (32) is computed for each of the simulation cases. The reaction rate for a porous medium with an embedded cavity of aspect ratios, $a = 4b$, $a = 2b$, $a \sim b$, located at the center of the medium are plotted in Fig. 8. The rates are normalized with respect to the reaction rate for the corresponding porous inclusion of the same shape. As shown in Fig. 8, for low and intermediate slip parameters, with the increase in the slip parameter, the effective reaction rate decreases for all the aspect

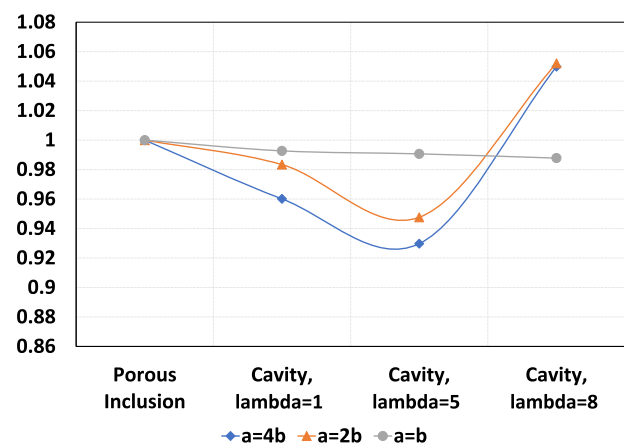


FIG. 8. Effective reaction rate computed at the outlet of a porous medium with an embedded cavity of different aspect ratios and slip parameters normalized with respect to the effective reaction rate for a porous inclusion of the same geometry, $a = 0.5$.

ratios. The slip parameter defines how the tangential velocity deviates from the cavity border. The relatively large slip parameter can result in highly bent streamlines, which limit the reactions on the sides while maintaining high reactivity in the fluid that enhances the downstream and overall reaction rates. The impact of the slip parameter on the effective reaction rate in the cavities with an aspect ratio closer to 1 is less important than the cavities with small aspect ratios. For a small slip parameter, the reaction rate in the domain with a fluid–fluid cavity is almost the same as, or slightly lower than that of, the case with a porous inclusion. In Fig. 9, the impact of the aspect ratio on the effective reaction rate is demonstrated. As is shown in Fig. 9, for a given value of a , the effective reaction rate decreases as b decreases. Although a narrow elongated cavity leaves more mineral content in the porous matrix available for dissolution, the

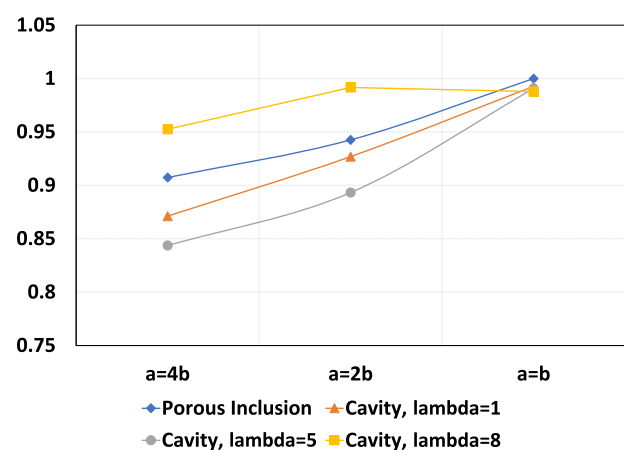


FIG. 9. Effective reaction rate computed at the outlet of a porous medium with an embedded cavity of different aspect ratios and slip parameters normalized with respect to the effective reaction rate for a spherical porous inclusion, $a = 0.5$.

cavity of a larger aspect ratio affects the fluid flow near the cavity significantly and enhances the dissolution. The effective reaction rate in the porous medium with a porous inclusion is observed to be more than the effective reaction rate in a medium with a fluid-filled cavity of the same shape. The results for severe slip on the cavity border do not show the same trend as low and medium slip and requires further investigation. For the slip parameter in the range of moderate and medium slip parameters, the effective reaction rate decreases as the slip parameter increases. The porous inclusion in most of the cases results in a higher effective reaction rate. The highest effective reaction rate corresponds to the medium with an embedded spherical porous inclusion. The results for severe slip on the cavity border did not show the same trend as low and medium slip and requires further investigation.

V. CONCLUSION

We performed a semi-analytical study of mineral dissolution in a porous matrix with embedded cavities. Dissolution of a calcite mineral in the vicinity and on the border of a fluid-filled cavity was compared to the one for a sediment-filled inclusion of the same geometry. Although the presented results in this study correspond to a single mineral dissolution, the proposed modeling framework can be easily adopted for multiple mineral dissolution. The cavity was considered to be a prolate spheroid with the longer axis in the direction of flow. Stokes flow and seepage flow (Darcy's law) were applied as two different physics governing the fluid flow in the cavity. In addition to the flow physics, the cavity aspect ratio and slip on the interface of the cavity and porous medium were investigated as the parameters that affect the dissolution.

Analytical solutions for flow and fluid velocity were derived for both fluid-filled cavities and sediment-filled inclusions. These results were used in a numerical simulator to solve the reactive transport process with a fully coupled built-in chemical reaction module.⁹⁰ The cavity volume may partially contribute to the overall flow as a result of the formation of vorticities. These features are not captured by Darcy's law, which is applied when the cavity is approximated by a porous sediment-filled inclusion.

Locating the dissolution hotspots is important in predicting the presume deformation of the cavity due to the dissolution. In the studied cases presented in this work, the dissolution hotspots in the porous matrix were mainly located at locations where fluid enters and leaves the cavity or inclusion. The thermodynamic driving forces, larger in those locations, are controlled by the overall advection pattern. However, local dissolution patterns, which are important for predicting the dynamic change of the cavity geometry, differ among the simulations. The sides of the cavity border in the upper parts showed minimum mineral dissolution for both porous inclusion and fluid-filled cavity. It was observed that the ultimate widening of the cavity due to dissolution could be expected at its very top and bottom.

The aspect ratio was observed to improve the total mineral dissolution at reaction hotspots. The loss in the mineral content at the upper and lower vertices of the spheroidal cavity (both fluid-filled and sediment-filled) increased when the aspect ratio approached unitary. This could be related to the impact of the cavity aspect

ratio on the fluid flow in the surrounding porous matrix and its contribution to enhancing the medium effective permeability.⁸⁹ The aspect ratio had no significant impact on the minimum cumulative loss in the mineral content along the border of the cavity. While, for an aspect ratio close to one, the spatial pattern of dissolution is similar between the fluid-filled cavity simulations and the ones with a porous medium inclusion assumption, regardless of the slip parameters used, observable differences exist for smaller aspect ratios. Overall, the local reaction rate at the borders on the sides is underestimated by the porous medium assumption. This difference is more evident with the smaller aspect ratio when an intermediate slip parameter is used. At the downstream of the cavity, the porous medium inclusion assumption tends to underestimate the dissolution in many cases when the aspect ratio is small or the slip parameter is large.

For a small slip parameter, the effective reaction rate in a porous medium with a sediment-filled inclusion was similar to the same medium with a fluid-filled inclusion. If the slip on the cavity-matrix interface is important, then the reaction rate in the medium with a porous inclusion will be larger than in the medium with a cavity. The discrepancy in the mineral loss on the boundaries of a porous inclusion compared to the boundaries of a fluid-filled cavity is more significant on the elongated cavities rather than the spherical cavities.

The effective reaction rate increased with the aspect ratio for a fixed value of a . The effective reaction rate in the medium with a porous inclusion assumption was observed to be more than the effective reaction rate in a medium with a fluid-filled cavity of the same shape. Overall, simulating a cavity as a highly permeable porous medium may result in the overestimation of the overall mineral dissolution and underestimation of local mineral dissolution in the area near the cavity.

Our study highlighted that simulating a cavity as a highly permeable zone may result in overestimation of the overall rock mineral dissolution while underestimating mineral dissolution locally, which would result in a shape that differs from the actual case in a dynamic case.

APPENDIX A: ORTHOGONAL CURVILINEAR COORDINATE SYSTEM

In a general orthogonal curvilinear coordinate system (χ_1, χ_2, χ_3) with $(\mathbf{e}_{\chi_1}, \mathbf{e}_{\chi_2}, \mathbf{e}_{\chi_3})$ as the base vectors and (h_1, h_2, h_3) as the corresponding scale factors, the del operator is given as

$$\nabla = \frac{1}{h_1} \frac{\partial}{\partial \chi_1} \mathbf{e}_{\chi_1} + \frac{1}{h_2} \frac{\partial}{\partial \chi_2} \mathbf{e}_{\chi_2} + \frac{1}{h_3} \frac{\partial}{\partial \chi_3} \mathbf{e}_{\chi_3}. \quad (\text{A1})$$

In the axis-symmetric system, $\frac{\partial}{\partial \chi_3} \mathbf{e}_{\chi_3} = 0$, the Laplacian ∇^2 operator and E^2 operator are as follows:

$$\nabla^2 = \frac{1}{h_1 h_2 h_3} \left[\frac{\partial}{\partial \chi_1} \left(\frac{h_2 h_3}{h_1} \frac{\partial}{\partial \chi_1} \right) + \frac{\partial}{\partial \chi_2} \left(\frac{h_1 h_3}{h_2} \frac{\partial}{\partial \chi_2} \right) \right] \quad (\text{A2})$$

and

$$E^2 = \frac{h_3}{h_1 h_2} \left[\frac{\partial}{\partial \chi_1} \left(\frac{h_2}{h_1 h_3} \frac{\partial}{\partial \chi_1} \right) + \frac{\partial}{\partial \chi_2} \left(\frac{h_1}{h_2 h_3} \frac{\partial}{\partial \chi_2} \right) \right]. \quad (\text{A3})$$

APPENDIX B: PROLATE SPHEROIDAL COORDINATE SYSTEM

The prolate spheroidal coordinate system (ξ, η, ϕ) with the normal unit vectors $(\mathbf{e}_\xi, \mathbf{e}_\eta, \mathbf{e}_\phi)$ relates to the Cartesian coordinate system through

$$\begin{aligned} x &= c \cosh \xi \cos \eta, \\ y &= c \sinh \xi \sin \eta \sin \phi, \\ z &= c \sinh \xi \sin \eta \cos \phi, \end{aligned} \quad (\text{B1})$$

with

$$0 \leq \xi < \infty, \quad 0 \leq \eta \leq \pi, \quad 0 \leq \phi \leq 2\pi, \quad (\text{B2})$$

and the focal length of the spheroid c as

$$c = \sqrt{a^2 - b^2}. \quad (\text{B3})$$

Usually, for the sake of convenience in notation, the following change of variables is commonly used:

$$s = \cosh \xi, \quad t = \cos \eta, \quad (\text{B4})$$

and then,

$$\begin{aligned} x &= cst, \\ y &= c\sqrt{s^2 - 1}\sqrt{1 - t^2} \sin \phi, \\ z &= c\sqrt{s^2 - 1}\sqrt{1 - t^2} \cos \phi, \end{aligned} \quad (\text{B5})$$

with

$$s \geq 1, \quad -1 \leq t \leq 1, \quad 0 \leq \phi \leq 2\pi. \quad (\text{B6})$$

The prolate spheroidal coordinate scale factors are given as

$$h_1 = h_2 = c\sqrt{s^2 - t^2}, \quad h_3 = c\sqrt{s^2 - 1}\sqrt{1 - t^2}. \quad (\text{B7})$$

Using (s, t, ϕ) as the new coordinate variables leads to the following differential equation:

$$\frac{\partial}{\partial \xi} = \sqrt{s^2 - 1} \frac{\partial}{\partial s}, \quad \frac{\partial}{\partial \eta} = -\sqrt{1 - t^2} \frac{\partial}{\partial t}. \quad (\text{B8})$$

The differential operator E^2 in the spheroidal coordinate system yields

$$E^2 = \frac{1}{c^2(s^2 - t^2)} \left[(s^2 - 1) \frac{\partial^2}{\partial s^2} + (1 - t^2) \frac{\partial^2}{\partial t^2} \right]. \quad (\text{B9})$$

The Laplacian differential operator for the spheroidal coordinate system is given as

$$\nabla^2 = \frac{1}{c^2(s^2 - t^2)} \left[(s^2 - 1) \frac{\partial^2}{\partial s^2} + (1 - t^2) \frac{\partial^2}{\partial t^2} + 2s \frac{\partial}{\partial s} - 2t \frac{\partial}{\partial t} \right]. \quad (\text{B10})$$

APPENDIX C: SOLUTION FOR FLOW

1. Fluid flow in the cavity

We apply the same method presented in by Ref. 89 to solve the axis-symmetric Stokes flow in the spheroidal coordinate system. We introduce the stream function Ψ_c inside the cavity as

$$\begin{aligned} u_c &= \frac{1}{h_2 h_3} \frac{\partial \Psi_c}{\partial \chi_2} \quad \text{in } \Omega_c, \\ v_c &= -\frac{1}{h_1 h_3} \frac{\partial \Psi_c}{\partial \chi_1} \quad \text{in } \Omega_c. \end{aligned} \quad (\text{C1})$$

Obtaining the curl of Eq. (5), together with some mathematical manipulations, the stream function in the Stokes media satisfies the following fourth order equation:

$$E^4 \Psi_c = E^2 (E^2 \Psi_c) = 0 \quad \text{in } \Omega_c, \quad (\text{C2})$$

where the differential operator E^2 is given in Eq. (A3). The pressure field inside the cavity obeys

$$\begin{aligned} -\frac{1}{h_1} \frac{\partial p_c}{\partial \chi_1} + \frac{\mu}{h_2 h_3} \frac{\partial}{\partial \chi_2} (E^2 \Psi_c) &= 0 \quad \text{in } \Omega_c, \\ -\frac{1}{h_2} \frac{\partial p_c}{\partial \chi_2} - \frac{\mu}{h_1 h_3} \frac{\partial}{\partial \chi_1} (E^2 \Psi_c) &= 0 \quad \text{in } \Omega_c. \end{aligned} \quad (\text{C3})$$

In the prolate spheroidal system, we introduce the following series expansion for the solution to Eq. (C2). According to the technique of the solution, the stream function Ψ_c is decomposed into two parts: the part already in $\text{Ker} E^2$ and the part not in $\text{Ker} E^2$ but still in $\text{Ker} E^4$, thus,

$$\begin{aligned} \Psi_c(s, t) &= \sum_{i=1}^4 (\Psi_c)_i(s, t) \\ &= c^2 \sum_{i=1}^4 \sum_{n=0}^{\infty} \Phi_{i,n}(s, t) + c^2 \sum_{i=1}^4 \sum_{n=0}^{\infty} a_{i,n} \Theta_{i,n}(s, t). \end{aligned} \quad (\text{C4})$$

Here, $a_{i,n}$ is a constant to be determined later, $\Theta_{i,n}(s, t)$ belongs to $\text{Ker} E^2$ and is a trivial solution for $\text{Ker} E^4$,

$$E^2 \Theta_{i,n} = 0, \quad E^4 \Theta_{i,n} = 0, \quad (\text{C5})$$

and $\Phi_{i,n}(s, t)$ represents the remainder of the functions in $\text{Ker} E^4$ that do not belong to $\text{Ker} E^2$,

$$E^2 \Phi_{i,n} \neq 0, \quad E^4 \Phi_{i,n} = 0. \quad (\text{C6})$$

The coefficient c^2 preceding either of the terms on the right-hand side of Eq. (C4) is for future convenience in formulation. $\Theta_{i,n}(s, t)$ is expressed as

$$\Theta_{1,n}(s) = G_n(s) G_n(t), \quad (\text{C7a})$$

$$\Theta_{2,n}(s) = G_n(s) H_n(t), \quad (\text{C7b})$$

$$\Theta_{3,n}(s) = H_n(s) G_n(t), \quad (\text{C7c})$$

$$\Theta_{4,n}(s) = H_n(s) H_n(t), \quad (\text{C7d})$$

where G_n and H_n are the Gegenbauer polynomials of the first and second kind and of degree $-\frac{1}{2}$.

The functions $\Phi_{i,n}(s, t)$ are defined so that

$$E^2 \Phi_{i,n}(s, t) = C_{i,n} \Theta_{i,n}(s, t) \quad \text{in } \Omega_c, \quad (\text{C8})$$

for $i = 1, \dots, 4, n = 0, 1, 2, \dots$, where $C_{i,n}$ is the constant to be defined later.

For $i = 1$, $\Phi_{1,n}$ (refer to Ref. 89 for more details) yields

$$\Phi_{1,n}(s, t) = c^2 C_{1,n} \left(\frac{-\alpha_n}{2(2n-1)} G_{n-2}(s) + \frac{\beta_n}{2(2n+1)} G_{n+2}(s) \right) G_n(t) - c^2 C_{1,n} G_n(s) \left(\frac{\alpha_n}{2(2n-1)} G_{n-2}(t) - \frac{\beta_n}{2(2n+1)} G_{n+2}(t) \right), \quad (C9)$$

where $C_{1,n}$ is a constant to be determined later and

$$\begin{aligned} \alpha_0 &= \alpha_1 = \alpha_2 = \alpha_3 = 0, \\ \gamma_0 &= 1, \quad \gamma_1 = 1, \quad \gamma_2 = \frac{1}{5}, \quad \gamma_3 = \frac{3}{7}, \\ \beta_0 &= -2, \quad \beta_1 = 2, \quad \beta_2 = \frac{4}{5}, \quad \beta_3 = \frac{4}{7}, \\ \alpha_n &= \frac{(n-3)(n-2)}{(2n-3)(2n-1)}, \quad n \geq 4 \\ \beta_n &= \frac{(n+1)(n+2)}{(2n-1)(2n+1)}, \quad n \geq 4 \\ \gamma_n &= \frac{2n^2 - 2n - 3}{(2n+1)(2n-3)}, \quad n \geq 4. \end{aligned} \quad (C10)$$

Therefore, $(\Psi_c)_1$ includes the complete set of the solution as follows:

$$(\Psi_c)_1(s, t) = c^2 \sum_{n=0}^{\infty} \Phi_{1,n}(s, t) + c^2 \sum_{n=0}^{\infty} a_{1,n} \Theta_{1,n}(s, t), \quad (C11)$$

which can be written as

$$(\Psi_c)_1(s, t) = c^2 \sum_{n=1}^{\infty} g_{1,n}(s) G_n(t), \quad (C12)$$

with $g_{1,n}$ given as

$$\begin{aligned} g_{1,n} &= a_{1,n} G_n(s) + c^2 \left(\frac{\beta_{n-2} C_{1,n-2}}{2(2n-3)} - \frac{\alpha_n C_{1,n}}{2(2n-1)} \right) G_{n-2}(s) \\ &+ c^2 \left(\frac{\beta_n C_{1,n}}{2(2n+1)} - \frac{\alpha_{n+2} C_{1,n+2}}{2(2n+3)} \right) G_{n+2}(s). \end{aligned} \quad (C13)$$

Applying the same procedure for $\Phi_{2,n}(s, t)$, $\Phi_{3,n}(s, t)$, $\Phi_{4,n}(s, t)$ in Eq. (C8), the complete set of solutions for the Stokes stream function can be determined. Here, the general solution for the Stokes stream function in the cavity can be simplified using the following assumptions:

- The solution has to be bounded on the axis of symmetry; thus, i is limited to $i = 1, 3$ in Eq. (C4), regarding the definition of $\Theta_{i,n}$ given in Eq. (C7).
- The solution has symmetry with respect to t , and then, only the even n values are used, $n = 2k$.
- The solution has to be bounded inside the spheroidal vug when $s \rightarrow 1$, and then, i is limited to $i = 1$, regarding the definition of $\Theta_{i,n}$ given in Eq. (C7).

Finally, the stream function inside the spheroidal cavity is given as

$$\Psi_c(s, t) = c^2 \sum_{k=1}^{\infty} g_{1,2k}(s) G_{2k}(t). \quad (C14)$$

Note that none of the individual terms in Eq. (C14) satisfies $E^4 \Psi_c(s, t) = 0$, but the sum of all the other terms do. This condition can be investigated using the method for developing the solution. The pressure field inside the cavity is related to stream functions using Eqs. (C3) and (B8) as follows:

$$\begin{aligned} (s^2 - 1) \frac{\partial p_c}{\partial s} &= -\frac{\mu}{c\rho} \frac{\partial}{\partial t} (E^2 \Psi_c), \\ (1 - t^2) \frac{\partial p_c}{\partial t} &= \frac{\mu}{c\rho} \frac{\partial}{\partial s} (E^2 \Psi_c). \end{aligned} \quad (C15)$$

APPENDIX D: COUPLING MATRIX AND STOKES MEDIUM

The condition given by Eq. (8) in spheroidal coordinates is as follows:

$$\frac{\partial \Psi_c}{\partial t} = \frac{\partial \Psi_m}{\partial t} \quad \text{on } s = s_0, \quad (D1)$$

which after associating the coefficients of the same t -dependent functions yields

$$g_{1,2k}(s_0) = -\frac{k_m}{2k(2k-1)\mu} (s_0^2 - 1) F'_{2k}(s_0). \quad (D2)$$

The generalized Beavers–Joseph–Saffman condition described in Eq. (10), with the tangential component of the rate of the strain tensor $\epsilon_{\xi\eta} = \epsilon_{\eta\xi}$ given as

$$\begin{aligned} \epsilon_{\xi\eta} &= \frac{1}{c^3} \left[\sqrt{\frac{(s^2 - 1)}{(1 - t^2)}} \left(\frac{-2s}{(s^2 - t^2)^2} \frac{\partial \Psi_c}{\partial s} + \frac{1}{(s^2 - t^2)} \frac{\partial^2 \Psi_c}{\partial s^2} \right) \right. \\ &\quad \left. - \sqrt{\frac{(1 - t^2)}{(s^2 - 1)}} \left(\frac{2t}{(s^2 - t^2)^2} \frac{\partial \Psi_c}{\partial t} + \frac{1}{(s^2 - t^2)} \frac{\partial^2 \Psi_c}{\partial t^2} \right) \right], \end{aligned} \quad (D3)$$

reduces to

$$\begin{aligned} (s^2 - t^2) \left(\frac{\partial^2 \Psi_c}{\partial s^2} - \frac{\partial^2 \Psi_c}{\partial t^2} \right) - 2s \frac{\partial \Psi_c}{\partial s} - 2t \frac{\partial \Psi_c}{\partial t} \\ = \frac{c\lambda}{\sqrt{k_m}} \sqrt{s^2 - 1} (s^2 - t^2)^{\frac{3}{2}} \frac{\partial \Psi_c}{\partial s} \quad \text{on } s = s_0. \end{aligned} \quad (D4)$$

Replacing the Stokes stream function from Eq. (C12) yields the following equation:

$$\begin{aligned} \sum_{k=1}^{\infty} -\frac{2s_0(s_0^2 - 1)}{2k(2k-1)} g'_{1,2k}(s_0) P'_{2k-1}(t) + 2g_{1,2k}(s_0) t P_{2k-1}(t) \\ + \left[\frac{s_0^2 - 1}{2k(2k-1)} g''_{1,2k}(s_0) + g_{1,2k}(s_0) \right] (s_0^2 - t^2) P'_{2k-1}(t) \\ = \sum_{k=1}^{\infty} \frac{\lambda c}{\sqrt{k_m}} \frac{\sqrt{s_0^2 - 1}}{2k(2k-1)} g'_{1,2k}(s_0) (s_0^2 - t^2)^{\frac{3}{2}} P'_{2k-1}(t). \end{aligned} \quad (D5)$$

It is complicated to separate the coefficients of $P_{2k-1}(t)$ or $P'_{2k-1}(t)$ for Eq. (D5) in order to equalize the coefficients of the same terms. We rewrite all the terms in the form of Legendre polynomial expansion series. For computational purposes, this is a good technique for working with dissimilar terms.

The pressure jump condition given by Eq. (9) in the spheroidal coordinate system with

$$\epsilon_{\xi\xi} = \frac{1}{c^3(s^2 - t^2)} \left(\frac{\partial^2 \Psi_c}{\partial s \partial t} + \frac{t}{s^2 - t^2} \frac{\partial \Psi_c}{\partial s} - \frac{s(2s^2 - 1 - t^2)}{(s^2 - t^2)(s^2 - 1)} \frac{\partial \Psi_c}{\partial t} \right) \quad (\text{D6})$$

is given as

$$\begin{aligned} & \sum_{k=1}^{\infty} \frac{c^2}{2} (C_{1,2k} P_{2k-1}(s_o) + F_{2k-1}(s)/\mu) (s_o^2 - t^2)^2 P_{2k-1}(t) \\ &= \sum_{k=1}^{\infty} g'_{1,2k}(s_o) (s_o^2 - t^2) P_{2k-1}(t) - \sum_{k=1}^{\infty} g'_{1,2k}(s_o) t G_{2k}(t) \\ & \quad - \sum_{k=1}^{\infty} g_{1,2k}(s_o) \frac{s_o(2s_o^2 - 1 - t^2)}{s_o^2 - 1} P_{2k-1}(t). \end{aligned} \quad (\text{D7})$$

We rewrite all of the terms in the form of the Legendre polynomial series expansion and equalize similar terms. For the sake of computational ease, we replace the infinite sum $\sum_{k=1}^{\infty}$ with finite sum $\sum_{k=1}^N$ with $N \geq 1$. Having $3N$ equations given by Eqs. (D2), (D5), and (D7), we can obtain $3N$ unknowns $C_{1,2k}$, $a_{1,2k}$, A_{2k} by solving an algebraic linear system of equations.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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